MODULE 2.3 — QUALITY OVERALL SUMMARY (QOS)

AMLOZA (Amlodipine Besylate Tablets, USP) 5 mg / 10 mg — ANDA

This Quality Overall Summary (QOS) provides an integrated summary of the chemistry, manufacturing, and controls (CMC) information presented in Module 3 for AMLOZA (amlodipine besylate) Tablets, 5 mg and 10 mg. It follows the ICH M4Q(R1) QOS format.

2.3.S DRUG SUBSTANCE (Amlodipine Besylate)

2.3.S.1 General Information

Amlodipine besylate is the besylate salt of amlodipine, a dihydropyridine calcium channel blocker. Chemical name: 3-Ethyl-5-methyl (±)-2-[(2-aminoethoxy)methyl]-4-(2-chlorophenyl)-1,4-dihydro-6-methyl-3,5-pyridinedicarboxylate,monobenzenesulphonate.

Molecular formula C20H25ClN2O5·C6H6O3S; molecular weight 567.1 g/mol. The drug substance is a white to off-white crystalline powder, sparingly soluble in water and slightly soluble in ethanol. Amlodipine has been reported in the literature as a highly soluble and highly permeable compound; this ANDA is supported by in vivo bioequivalence rather than a BCS-based biowaiver.

2.3.S.2 Manufacture

Amlodipine besylate is sourced from a qualified Drug Master File (DMF) holder. [DMF number to be referenced upon selection of API supplier — placeholder for academic exercise]. The synthetic route, in-process controls, and manufacturing process description are held in the API DMF and are incorporated by reference; this ANDA relies on the DMF holder's Letter of Authorization (LOA).

2.3.S.3 Characterisation

Structure elucidation is supported by IR, NMR, mass spectrometry, and elemental analysis data held in the DMF. The supplied API is controlled for the approved solid-state (polymorphic) form as described in the referenced DMF, with polymorph identity confirmed via X-ray powder diffraction at drug substance release. Potential process-related impurities and degradation products are controlled per ICH Q3A/Q3B thresholds.

2.3.S.4 Control of Drug Substance

Specification basis	USP monograph for Amlodipine Besylate, supplemented by in-house limits for process-related impurities not covered compendially
Key tests	Description, Identification (IR/HPLC), Assay (HPLC, 98.0–102.0% on anhydrous basis), Related Substances (individual and total impurities by HPLC), Water Content (KF), Residual Solvents (ICH Q3C), Elemental Impurities (ICH Q3D / USP <232>/<233>), Residue on Ignition, Particle Size Distribution (development/process parameter)
Analytical methods	Compendial (USP) methods used where available; any in-house methods to be validated per ICH Q2(R1) prior to use

2.3.S.5 Reference Standards

USP Reference Standard for Amlodipine Besylate is used for identification and assay. Working standards are qualified against the USP standard prior to use.

2.3.S.6 Container Closure System

The drug substance is packaged in double polyethylene bags within a fiber drum, or equivalent, consistent with DMF specifications, providing protection from moisture and light.

2.3.S.7 Stability

A stability program per ICH Q1A(R2) is proposed, comprising long-term (25°C/60% RH) and accelerated (40°C/75% RH) storage conditions. No experimental stability data are available or presented in this academic dossier; the retest period would be established from data generated by the DMF holder under this or an equivalent approved protocol.

2.3.P DRUG PRODUCT (AMLOZA Tablets, 5 mg / 10 mg)

2.3.P.1 Description and Composition

AMLOZA is formulated as an uncoated immediate-release tablet, white, round, biconvex, for oral administration. Each tablet contains amlodipine besylate equivalent to 5 mg or 10 mg amlodipine base.

Active ingredient	Amlodipine Besylate USP (eq. to 5 mg / 10 mg amlodipine)
Diluent	Microcrystalline Cellulose NF
Diluent/Binder	Anhydrous Dibasic Calcium Phosphate USP
Disintegrant	Sodium Starch Glycolate, Type A NF
Lubricant	Magnesium Stearate NF

2.3.P.2 Pharmaceutical Development

The formulation was developed to be Q1 same and Q2 similar relative to the Reference Listed Drug (Norvasc Tablets), consistent with FDA expectations for generic immediate-release oral solid dosage forms, using standard, compendial excipients with no known absorption-modifying interactions with amlodipine. Amlodipine has been reported in the literature as a highly soluble and highly permeable compound; this ANDA is supported by in vivo bioequivalence rather than a BCS-based biowaiver. Direct compression was selected as the manufacturing process, as development studies supported the suitability of the drug substance's flow and compressibility characteristics for this route, avoiding unnecessary processing steps (wet granulation) that could introduce degradation risk given amlodipine's known sensitivity to certain processing conditions.

2.3.P.3 Manufacture

Manufacturing process	Direct compression: dispensing → sifting/blending of drug substance with diluents and disintegrant → lubrication → compression → packaging
Batch formula basis	Representative exhibit batch example: 100,000 tablets (10 mg strength, 20.000 kg total batch weight), consistent with FDA's minimum exhibit batch size recommendation for solid oral dosage forms; see Module 3.2.P.3.2 for the full quantitative batch formula
Critical process parameters	Blending time/speed, compression force (tablet hardness/friability target ranges), machine speed
In-process controls	Blend uniformity, tablet weight variation, hardness, friability, thickness, disintegration time

2.3.P.4 Control of Excipients

All excipients comply with their respective NF/USP compendial monographs. No novel or non-compendial excipients are used in this formulation.

2.3.P.5 Control of Drug Product

Specification basis	USP monograph for Amlodipine Tablets, supplemented by in-house limits
Key release tests	Description, Identification, Assay (95.0–105.0%), Content Uniformity (USP <905>), Dissolution (USP apparatus and medium per compendial method or product-specific FDA guidance), Related Substances, Water Content, Microbial Limits (if applicable)
Dissolution basis	Comparative dissolution characterization is proposed to support biopharmaceutic comparability against the RLD (Norvasc) across pH 1.2, 4.5, and 6.8 media, using a method to be qualified during formal development (see Module 3.2.P.5.1 and Module 5 for an example method); results feed the overall bioequivalence narrative (see Module 5)

2.3.P.6 Reference Standards

As described in 2.3.S.5; the same USP Reference Standard is used for drug product testing.

2.3.P.7 Container Closure System

Primary packaging: high-density polyethylene (HDPE) bottles with child-resistant closures, and/or PVC/aluminium blister strips, consistent with RLD presentation. Container closure system suitability supported by compendial compliance and compatibility assessment.

2.3.P.8 Stability

A stability program per ICH Q1A(R2) is proposed: long-term storage at 25°C ± 2°C / 60% RH ± 5% RH, and accelerated storage at 40°C ± 2°C / 75% RH ± 5% RH, with testing proposed at 0, 3, 6, 9, 12, 18, and 24 months (long-term) and 0, 3, and 6 months (accelerated). No experimental stability data are available or presented in this academic dossier; shelf life would be established from data generated under this or an equivalent approved protocol. Proposed storage statement: 'Store at 25°C (77°F); excursions permitted between 15–30°C (59–86°F),' consistent with the RLD labelling.

Portfolio Note: This QOS is written to demonstrate understanding of ICH M4Q(R1) structure and how Module 2.3 content is derived from and summarizes Module 3 data. Bracketed placeholders (DMF number, batch size, retest period) reflect data that would come from actual manufacturing/DMF sources in a real submission and are not fabricated as if real.